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2 **Supporting Information for**

3 **Rational Quantum Mechanics: Testing Quantum Theory with Quantum Computers**

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7 **This PDF file includes:**

8 Supporting text

9 Figs. S1 to S3

10 SI References

11 Supporting Information Text

12 The primary purpose of the SI is to describe explicitly the construction of the discretised Riemann Sphere, on which the
 13 discretised Hilbert Space of RaQM is based. Non-trivial arithmetic properties of the discretised Riemann Sphere are derived.
 14 A constructive derivation of the Uncertainty Principle is given.

15 Discretising the Riemann Sphere

16 The continuum field $\mathbb{C}$, with its axiomatically defined element $i = \sqrt{-1}$, plays an essential role in QM (either explicitly or
 17 implicitly as in the de Broglie-Bohm interpretation). Central to RaQM is a discretisation of the Riemann Sphere $\mathbb{C} \cup \{\infty\}$ of
 18 (extended) complex numbers $z(\theta, \phi) = \cot(\theta/2)e^{i\phi}$ where θ denotes co-latitude (or zenith angle) and ϕ denotes longitude (or
 19 azimuth). The discretisation is defined by

$$20 \quad \cos^2 \frac{\theta}{2} = \frac{m}{L} \in \mathbb{Q}; \quad \frac{\phi}{2\pi} = \frac{n}{L} \in \mathbb{Q}. \quad [1]$$

21 where $L \in \mathbb{N}$ and $0 \leq m, n \leq L$ are whole numbers. Note that if θ satisfies (1) then since $\cos \theta = 2 \cos^2(\theta/2) - 1$, necessarily
 22 $\cos \theta \in \mathbb{Q}$. L is an especially important variable, defining the degree of granularity of discretised Hilbert Space: QM is the
 23 (singular) limit of RaQM at $L = \infty$, whilst the classical limit of RaQM occurs at $L = 1$. The key property of this discretisation is
 24 that the i (with property $i^2 = -1$) does not have to be introduced by axiom. Rather, as discussed below, i and the corresponding
 25 quaternions are defined constructively from representations of complex numbers as permutation/negation operators acting on
 26 bit strings. With this definition of i , the entire set $\{z(\theta, \phi)\}$ of complex numbers on the discretised Riemann Sphere is obtained
 27 constructively by an inductive argument, for all $L \in \mathbb{N}$.

28 **Discretised Complex Numbers and Quaternions.** When $L = 2^0 = 1$, (1) implies $\theta \in \{0, \pi\}$, $\phi = 0$, i.e., the discretisation only
 29 admits 2 points at the north and south poles. We represent the corresponding set of ‘complex numbers’ in terms of the identity
 30 and negation operators

$$31 \quad z(0, 0)\{1\} = \{1\}; \quad z(\pi, 0)\{1\} = \{-1\} = -\{1\} \quad [2]$$

32 With $L = 2^0$ (the classical limit of RaQM), the discretisation is too coarse to admit complex structure at all. Complex numbers
 33 play no essential role in classical physics.

34 When $L = 2^1$, then $\theta \in \{0, \pi/2, \pi\}$, $\phi \in \{0, \pi\}$ according to (1). $L = 2$ is large enough to admit complex structure, represented
 35 by the permutation/negation operator (PNO) i defined as

$$36 \quad i\{a_1, a_2\} = \{-a_2, a_1\} \quad [3]$$

where $a_i \in \{1, -1\}$, so that $i^2\{a_1, a_2\} = \{-a_1, -a_2\} \equiv -\{a_1, a_2\}$. Generalising (2), the set of complex numbers on the $L = 2$
 discretised Riemann Sphere are associated with the 4 PNOs $\{i^0, i, i^2, i^3\}$. Specifically,

$$\begin{aligned} z(0, 0)\{1, 1\} &= i^0\{1, 1\} = \{1, 1\}; \\ z\left(\frac{\pi}{2}, 0\right)\{1, 1\} &= i\{1, 1\} = \{-1, 1\}; \\ z(\pi, 0)\{1, 1\} &= z(\pi, \pi)\{1, 1\} = i^2\{1, 1\} = \{-1, -1\}; \\ z\left(\frac{\pi}{2}, \pi\right)\{1, 1\} &= i^3\{1, 1\} = \{1, -1\} \end{aligned} \quad [4]$$

37 as shown in Fig S1a.

When $L = 2^2 = 4$ then $\theta \in \{0, \pi/3, \pi/2, 2\pi/3, \pi\}$, $\phi \in \{0, \pi/2, \pi, 3\pi/2\}$ according to (1). The discretised sphere now comprises
 8 points on each of two complete great circles (at $\phi = 0, \pi$ and $\phi = \pi/2, 3\pi/2$) passing through the north and south poles - 14
 points in total. $L = 4$ is now large enough to admit quaternionic structure, as shown below. The complex numbers at these
 points are represented as PNOs acting on 4-bit strings. We start by considering the points on the great circle $\phi = 0, \pi$, writing
 $\{1, 1, 1, 1\}$ as two concatenated 2-bit strings $\{1, 1\}|\{1, 1\}$. We define PNOs based on $\{i^0, i, i^2, i^3\}$ acting individually on each pair
 of 2-bit strings, with a construction motivated by the theory of spinors. Specifically, we first apply i twice to the second 2-bit
 string keeping the first fixed. We then apply $-i$ twice to the first 2-bit string keeping the second fixed; then again apply i twice
 to the second bit string keeping the first fixed; then again apply $-i$ twice to the first bit string keeping the second fixed. In
 total, from Fig S1a, the first 2-bit string has been rotated by 4π relative to the second (like Dirac scissors relative to the back

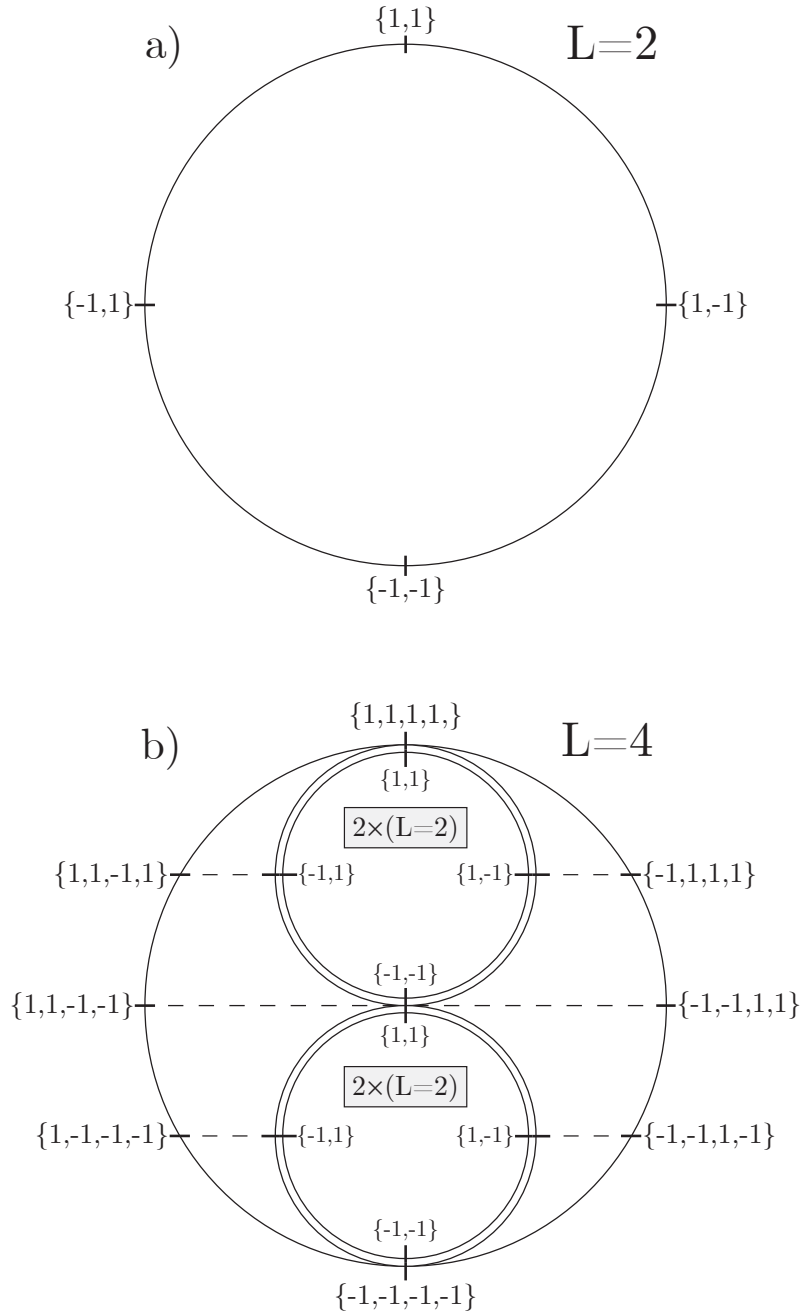


Fig. S1. a) The discretised Riemann Sphere at $L = 2$, represented as four 2-bit strings on a complete great circle at $\phi = 0, \pi$ through the poles. b) A schematic illustration of how to construct the discretised Riemann Sphere for $L = 4$, again on the great circle through $\phi = 0, \pi$, from two $L = 2$ great circles as in a). See the text for details. This also illustrates the construction of an $L = 2^M$ discretised Riemann Sphere on the great circle at $\phi = 0, \pi$, from two $L/2$ -bit discretised great circles.

of the chair). Specifically,

$$\begin{aligned}
z(0, 0)\{1, 1, 1, 1\} &= i^0\{1, 1\}|i^0\{1, 1\} = \{1, 1, 1, 1\}; \\
z(\frac{\pi}{3}, 0)\{1, 1, 1, 1\} &= i^0\{1, 1\}|i^1\{1, 1\} = \{1, 1, -1, 1\}; \\
z(\frac{\pi}{2}, 0)\{1, 1, 1, 1\} &= i^0\{1, 1\}|i^2\{1, 1\} = \{1, 1, -1, -1\}; \\
z(\frac{2\pi}{3}, 0)\{1, 1, 1, 1\} &= i^3\{1, 1\}|i^2\{1, 1\} = \{1, -1, -1, -1\}; \\
z(\pi, 0)\{1, 1, 1, 1\} &= z(\pi, \pi)\{1, 1, 1, 1\} = i^2\{1, 1\}|i^2\{1, 1\} = \{-1, -1, -1, -1\}; \\
z(\frac{2\pi}{3}, \pi)\{1, 1, 1, 1\} &= i^2\{1, 1\}|i^3\{1, 1\} = \{-1, -1, 1, -1\}; \\
z(\frac{\pi}{2}, \pi)\{1, 1, 1, 1\} &= i^2\{1, 1\}|i^0\{1, 1\} = \{-1, -1, 1, 1\}; \\
z(\frac{\pi}{3}, \pi)\{1, 1, 1, 1\} &= i^1\{1, 1\}|i^0\{1, 1\} = \{-1, 1, 1, 1\}
\end{aligned} \tag{5}$$

This construction is illustrated schematically in Fig S1b. Here the 4-bit strings on the $L = 4$ great circle comprise two 2-bit strings from the pair of smaller $L = 2$ circles, where one rotates in steps of $\pi/2$ by 4π relative to the other.

The complex numbers on the $\phi = \pi/2, 3\pi/2$ great circle are PNOs obtained by a cyclic permutation of bit strings on the $\phi = 0, \pi$ great circle, so that

$$\begin{aligned}
z(0, \frac{\pi}{2})\{1, 1, 1, 1\} &= \{1, 1, 1, 1\}; \\
z(\frac{\pi}{3}, \frac{\pi}{2})\{1, 1, 1, 1\} &= \{1, -1, 1, 1\}; \\
z(\frac{\pi}{2}, \frac{\pi}{2})\{1, 1, 1, 1\} &= \{1, -1, -1, 1\}; \\
z(\frac{2\pi}{3}, \frac{\pi}{2})\{1, 1, 1, 1\} &= \{-1, -1, -1, 1\}; \\
z(\pi, \frac{\pi}{2})\{1, 1, 1, 1\} &= z(\pi, \frac{3\pi}{2}) = \{-1, -1, -1, -1\}; \\
z(\frac{2\pi}{3}, \frac{3\pi}{2})\{1, 1, 1, 1\} &= \{-1, 1, -1, -1\}; \\
z(\frac{\pi}{2}, \frac{3\pi}{2})\{1, 1, 1, 1\} &= \{-1, 1, 1, -1\}; \\
z(\frac{\pi}{3}, \frac{3\pi}{2})\{1, 1, 1, 1\} &= \{1, 1, 1, -1\}.
\end{aligned} \tag{6}$$

That this cyclic permutation is consistent with quaternionic structure can be seen by defining the operators

$$\begin{aligned}
I\{a_1, a_2, a_3, a_4\} &= \{a_3, a_4, -a_1, -a_2\} \\
J\{a_1, a_2, a_3, a_4\} &= \{a_2, -a_1, -a_4, a_3\} \\
K\{a_1, a_2, a_3, a_4\} &= \{-a_4, a_3, -a_2, a_1\}
\end{aligned} \tag{7}$$

which satisfy

$$I^2 = J^2 = K^2 = -1; \quad I \times J = K. \tag{8}$$

In particular, $I\{1, 1, 1, 1\} = \{1, 1, -1, -1\} = z(\pi/2, 0)\{1, 1, 1, 1\}$ and $J\{1, 1, 1, 1\} = \{1, -1, -1, 1\} = z(\pi/2, \pi/2)\{1, 1, 1, 1\}$, consistent with I and J rotating the 4-bit string $\{1, 1, 1, 1\}$ at the north pole to the 4-bit strings on the equator at $\phi = 0$ and $\phi = \pi/2$, respectively. Consistent with this, K maps one these equatorial points to the other:

$$KI\{1, 1, 1, 1\} = K\{1, 1, -1, -1\} = \{1, -1, -1, 1\} = J\{1, 1, 1, 1\}. \tag{9}$$

The 14 4-bit strings on the 3D $L = 4$ discretised Riemann sphere are shown in Fig S2. The inductive step is to write the discretised PNOs for some $L = 2^M$, acting on the L -bit string $\{1, 1, \dots, 1\}$ as PNOs acting on 2 concatenated $L/2$ -bit strings $\{1, 1, \dots, 1\}|\{1, 1, \dots, 1\}$. The PNOs on the $\phi = 0, \pi$ great circle are defined using the spinorial construction illustrated in Fig S2b, generalised so that the two smaller circles represent the two $L/2$ discretised great circles at $\phi = 0, \pi$. We then rotate the $\phi = 0, \pi$ great circle around the polar axis by cyclically permuting the L -bit string as in

$$\zeta\{a_1, a_2, \dots, a_L\} \mapsto \{a_2, a_3, \dots, a_L, a_1\} \tag{10}$$

Each application of ζ corresponds to a rotation by $2\pi/L$ radians.

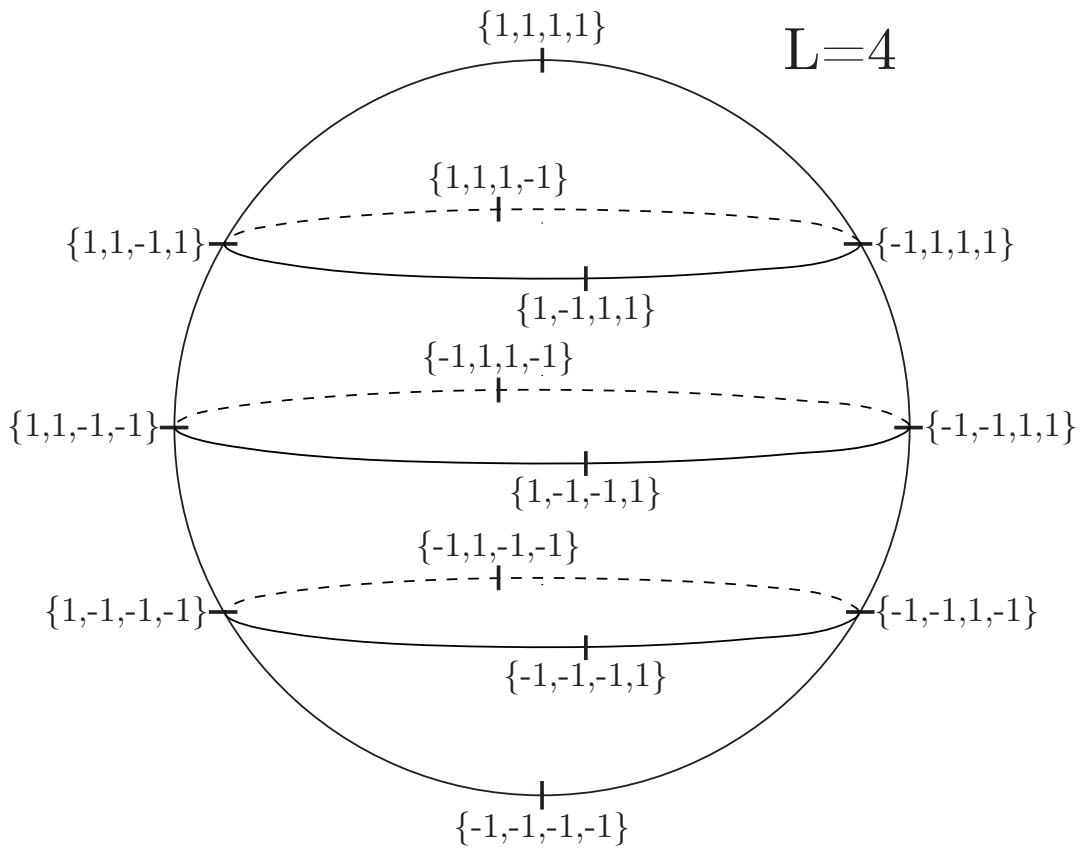


Fig. S2. The full 3D discretised Riemann Sphere for $L = 4$ - where, consistent with quaternionic structure (see text), rotation about the polar axes corresponding to a cyclic permutation of the 4-bit strings.

This completes the L -discretisation of the Riemann Sphere when L is a power of 2. For discretisations where L is not a power of 2, one can simply interpolate between constructions where L is a power of 2. For example, with $L = 3$, one can simply remove the last bit from the $L = 4$ construction on $\phi = 0$, yielding

$$\begin{aligned} z(0,0)\{1,1,1\} &= \{1,1,1\}; \\ z(\theta_*,0)\{1,1,1\} &= \{1,1,-1\}; \\ z(\pi-\theta_*,0)\{1,1,1\} &= \{1,-1,-1\}; \\ z(\pi,0)\{1,1,1\} &= \{-1,-1,-1\} \end{aligned} \quad [11]$$

where $\cos \theta_* = 1/3$. For the longitudes $\phi = 2\pi/3, 4\pi/3$, one simply performs two cyclic permutations to the 3-bit strings.

Note that, for all L , the proportion of 1s in the bit-string representing $z(\theta, \phi)$ is equal to $\cos^2 \theta/2$. Hence the proportion of -1s equals $\sin^2 \theta/2$, and the relative frequency of 1s to -1s is equal to $\cot^2 \theta/2$. This is the basis for the claim that there is no need to postulate Born's Rule in RaQM - it is redundant in the Dirac-von Neumann axioms.

Arithmetic Properties of Complex Numbers on the Discretised Riemann Sphere. Central to the physical interpretation of RaQM are the arithmetic properties of the complex-number PNOs on the L -bit discretisation of the Riemann Sphere. In particular, suppose z_1 and z_2 are complex numbers belonging to the L -discretised sphere. The key question we consider in this section is whether $z_1 \pm z_2$ belongs to the discretised sphere. We show that, typically, but non-trivially, they do not.

Consider, for example, $z_1 - z_2$. On the Argand plane, z_1 , z_2 and $z_2 - z_1$ correspond to three sides of a triangle with vertices at the origin, at z_1 and at z_2 . We will assume the triangle is non-degenerate, i.e. the angle ϕ between z_1 and z_2 is non-zero. On the Riemann Sphere the corresponding triangle is a spherical triangle with vertices which we label O , 1 and 2 (where O corresponds to the south pole).

Now if $z_1 = \cot(\theta_1/2)e^{i\phi_1}$ and $z_2 = \cot(\theta_2/2)e^{i\phi_2}$, both satisfy the rationality conditions (1) then $\cos \theta_1 \in \mathbb{Q}$, $\cos \theta_2 \in \mathbb{Q}$ and $\phi/2\pi = (\phi_2 - \phi_1)/2\pi \in \mathbb{Q}$. If we write $z_2 - z_1 = \cot(\theta_3/2)e^{i\phi_3}$, then for $z_2 - z_1$ to lie on the discretised Riemann Sphere, we require that $\cos \theta_3 \in \mathbb{Q}$.

The cosine rule applied to the triangle $\triangle O12$ can be written

$$\cos \theta_3 = \cos \theta_1 \cos \theta_2 + \sin \theta_1 \sin \theta_2 \cos \phi \quad [12]$$

If $\cos \theta_3 \in \mathbb{Q}$, then, since the first term on the RHS of (12) is rational, the second term on the RHS must also be rational. Squaring, this implies that

$$(1 - \cos^2 \theta_1)(1 - \cos^2 \theta_2) \cos^2 \phi \in \mathbb{Q} \quad [13]$$

and hence $\cos^2 \phi \in \mathbb{Q}$ and therefore $\cos 2\phi \in \mathbb{Q}$. However, we now have a contradiction since we know that $\phi/2\pi \in \mathbb{Q}$. According to Niven's Theorem (1) (2), the only values $0 \leq \phi < 2\pi$ where $\cos 2\phi$ and $2\phi/2\pi$ are simultaneously rational are:

$$2\phi = 0, \frac{\pi}{3}, \frac{\pi}{2}, \frac{2\pi}{3}, \pi, \frac{4\pi}{3}, \frac{3\pi}{2}, \frac{5\pi}{3}. \quad [14]$$

Hence, unless 2ϕ is a multiple of 60° *exactly*, $\cos 2\phi$ and $2\phi/2\pi$ cannot be simultaneously rational. Hence, modulo these exceptions, if z_1 and z_2 satisfy (1), $z_2 - z_1$ cannot satisfy (1). Of course, the same conclusion can be drawn if we add z_2 to z_1 . In the argument above we merely replace ϕ with $\pi - \phi$.

We refer to this result as the 'Impossible Triangle Corollary' (to Niven's Theorem). Niven's Theorem is central to RaQM and plays a vital role in the interpretation of complementarity, non-commuting observables and to the violation of Bell inequalities in RaQM. In these interpretations the mathematically impossible bases (one with irrational squared amplitudes and/or phases) correspond to physically impossible simultaneous counterfactual measurements.

Uncertainty Principle

In RaQM, the Uncertainty Principle for a single qubit arises from a trigonometric inequality on the sphere, together with Niven's Theorem. We then discuss the Uncertainty Principle for multiple qubits.

Consider a point p on the unit sphere (Fig S3) whose colatitude with respect to the three orthogonal poles p_x , p_y and p_z is θ , θ' and θ'' respectively. The internal angles ϕ' and η are shown on the figure. By the sine rule for spherical triangle $\triangle pp_xp_y$

$$\frac{\sin \theta''}{\sin \phi'} = \frac{\sin \pi/2}{\sin \eta} = \frac{1}{\sin \eta} \quad [15]$$

Hence

$$|\sin \theta''| \geq |\sin \phi'| \quad [16]$$

By the cosine rule for spherical triangle $\triangle pp_xp_z$,

$$\cos \theta = \sin \theta' \sin \phi' \quad [17]$$

From (16)

$$|\sin \theta'| |\sin \theta''| \geq |\sin \theta'| |\sin \phi'| \quad [18]$$

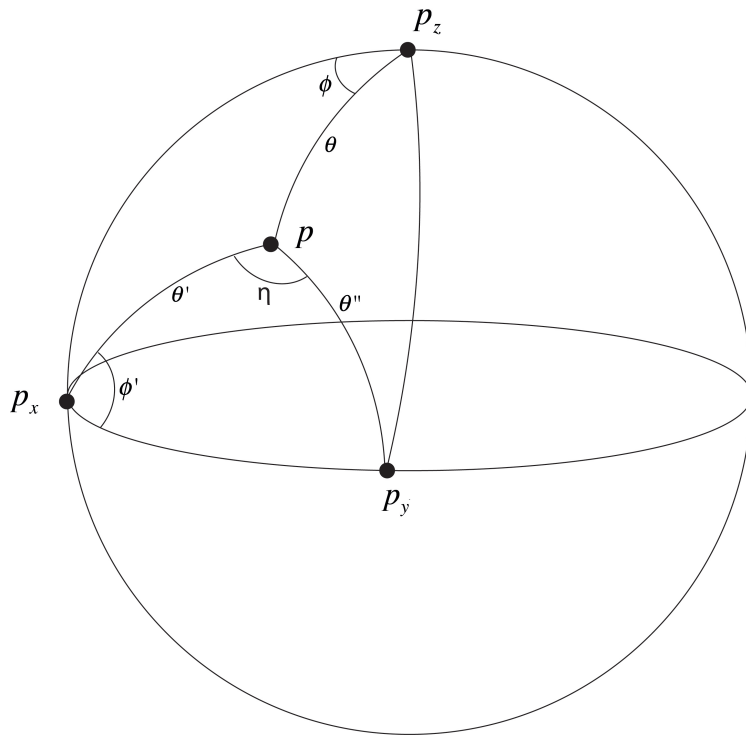


Fig. S3. In RaQM, the Uncertainty Principle $\Delta S_x \Delta S_y \geq \frac{\hbar}{2} \overline{S_z}$ for a single spin qubit arises from the trigonometry of spherical triangles and the correspondence between points on the discretised Bloch sphere with coordinates satisfying the rationality and bit strings.

95 and using (17)

$$96 \quad |\sin \theta'| |\sin \theta''| \geq |\cos \theta| \quad [19]$$

97 As discussed, a bit string at colatitude θ has a mean value $\mu_\theta = \cos \theta$, and standard deviation $\sigma_\theta = \sin \theta$. With this in mind,
 98 consider three discretised Bloch spheres, with the north poles oriented at p_x , p_y and p_z respectively. With $\cos \theta = \mu_\theta$, $\sin \theta = \sigma_\theta$
 99 (the mean and standard deviation of the bit string) then from (19),

$$100 \quad \sigma_{\theta'} \sigma_{\theta''} \geq |\mu_\theta| \quad [20]$$

101 If instead of ± 1 , the bit strings have dimensional values $\pm \hbar/2$ in order that they correspond to physical spin, then (20) becomes
 102 the familiar uncertainty principle for spin qubits

$$103 \quad \Delta S_x \Delta S_y \geq \frac{\hbar}{2} \bar{S}_z \quad [21]$$

104 The rationality constraints play a key role here. If, for example, $\cos \theta \in \mathbb{Q}$ and $\phi \in \mathbb{Q}$, by Niven's Theorem applied to
 105 $\Delta p p_x p_z$, $\cos \theta'$ cannot be rational. Hence it is impossible to know simultaneously, the spin values of a particle with respect to
 106 the two directions p_x and p_z (similarly for any two other pairs of directions).

107 Let us now derive the Uncertainty Principle for conjugate observables defined on multiple qubits (like position and
 108 momentum). First we square (20) and average over M points p on the discretised Bloch Sphere. Then, from (19),

$$109 \quad \overline{\sigma_{\theta'}^2 \sigma_{\theta''}^2} \geq \overline{|\mu_\theta|^2} \quad [22]$$

Now

$$\begin{aligned} \overline{\sigma_{\theta'}^2 \sigma_{\theta''}^2} &= \frac{1}{M} (\sigma_{\theta'_1}^2 + \sigma_{\theta'_2}^2 + \dots + \sigma_{\theta'_M}^2) \times \frac{1}{M} (\sigma_{\theta''_1}^2 + \sigma_{\theta''_2}^2 + \dots + \sigma_{\theta''_M}^2) \\ &> \frac{1}{M} ((\sigma_{\theta'_1}^2 \sigma_{\theta''_1}^2 + \sigma_{\theta'_2}^2 \sigma_{\theta''_2}^2 + \dots + \sigma_{\theta'_M}^2 \sigma_{\theta''_M}^2)) = \overline{\sigma_{\theta'}^2 \sigma_{\theta''}^2} \end{aligned} \quad [23]$$

110 Hence, if the M points are uniformly distributed with respect to $\cos \theta$ so that $\overline{|\cos \theta|} = 1/2$, then

$$111 \quad \sqrt{\overline{\sigma_{\theta'}^2}} \sqrt{\overline{\sigma_{\theta''}^2}} \geq \frac{1}{2} \quad [24]$$

112 Multiplying by $\hbar$ gives the standard position/momentum form

$$113 \quad \Delta x \Delta p \geq \frac{1}{2} \hbar \quad [25]$$

114 of the Uncertainty Principle.

115 References

- 116 1. J Jahnel, When does the (co)-sine of a rational angle give a rational number? (arXiv:1006.2938) (2010).
- 117 2. I Niven, *Irrational Numbers*. (The Mathematical Association of America), (1956).